Bharati Peddinti

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SUMMARY

AI and **Machine Learning Research Scientist** specializing in NLP, deep learning, and medical AI, with a track record of deploying impactful models in healthcare. Published author in peer-reviewed journals (IEEE, Elsevier) and contributor to state-of-the-art solutions in conversational AI, medical summarization, and computer vision. Proficient in PyTorch, TensorFlow, and foundation models, with expertise in translating research into production-grade systems. Passionate about advancing medical NLP to improve clinician workflows and patient outcomes.

TECHNICAL SKILLS

- **Artificial Intelligence & Machine Learning**: Data Cleaning, Regression, Decision Trees, Random Forest, Bagging & Boosting, Deep Learning (CNN, LSTMs), LLMs, Clustering, NLP, Azure ML, AutoML, MLOps CI/CD, Model Risk Management, Model Validations, Machine Learning
- **Cloud & Containerization Technologies**: AWS, AWS Sagemaker, Teradata, Docker, Kubernetes, Azure ML, Azure AutoML, GCP, Data Systems
- **Explainable AI**: SHAPLEY, Tree Explainer, XGBoost, Explainable AI (XAI)
- **Computer Vision**: Computer Vision(OpenCV, YOLOv8), OCR, Image Processing, Text Analytics, Content Moderation, Hugging Face
- **Natural Language Processing & Conversational AI**: Semantic Analysis, REST API, Bot Framework, Bot Emulator, Generative AI (Claude, LlamaIndex, Grok, Anthropic, Gemini), Prompt Engineering
- **Statistical Analysis & Predictive Modeling**: Predictive Analytics, Hypothesis Testing, PCA, Dimensionality Reduction, Stata, SAS
- **Programming Languages & Frameworks**: Python (Flask, FastAPI, NumPy, Pandas, Scikit-learn, Django, TensorFlow, Keras, PyTorch, Streamlit, Tkinter), SQL, Generative AI, ElasticSearch, SQLite, REST APIs
- **Development Tools**: Jupyter Notebook, Google Colab, PyCharm, Apache Spark, Tableau, R, Minitab, MS Excel, Power BI, Git, GitHub, GRC Archer, Jira, Service Now
- **Quantitative & Domain Expertise**: Economics, Statistics, Quantitative Analysis, Computer Science, Data Science

PROFESSIONAL EXPERIENCE

Armada IQ — **Jan 2025 - Present**
AI/ML Engineer — *Charlotte, NC*

- Developed end-to-end machine learning solutions by mining and extracting data, integrating **SQL** for dynamic filtering and semantic similarity scoring to enhance candidate-job alignment, and applying statistical techniques to drive insights.
- Engineered **CI/CD** pipelines integrating **LLMs (Claude, Grok, LLaMAIndex)** for resume and job description processing, dynamically rephrasing content and mapping key skills with job requirements using **HuggingFace** transformers and generative text synthesis.
- Implemented OCR and computer vision workflows using **Tesseract and OpenCV** to extract text from unstructured documents, reinforcing data quality and compliance with PySpark-based validation checks.
- Created **LangChain**-powered conversational **AI agents** for event bookings, integrating Anthropic Claude with calendar APIs and payment gateways to automate multi-turn dialogue management and resolve scheduling conflicts.
- Developed a pipeline for surgical video analysis using **OpenCV** to extract frames, **YOLOv8** for surgical tool detection, and segmentation models (MedSam, SAM, SAM2) to generate precise masks for client-specific medical datasets.
- Fine-tuned **MobileSAM** and **SAM2** on custom surgical datasets, enhancing mask accuracy for critical areas and aligning with client requirements for medical training and analysis.
- Synthesized data and augmented images and videos through Generative AI and prompt engineering to streamline automation processes and improve overall data quality.
- Containerized **TensorFlow and PyTorch** models using **Docker** and deployed scalable microservices via **Kubernetes,** enabling low-la- tency inference through **REST APIs** in enterprise environments.
- Automated **MLOps** pipelines using AWS SageMaker for model retraining and monitoring, integrating GitHub Actions to maintain performance and scalability across production environments.
- Deployed scalable web services using **Flask, FastAPI, and RESTful APIs** to seamlessly integrate AI models into enterprise applications, and created visualizations with Tableau to support data-driven decision-making.

Flagstar Bank N.A — **May 2024 - Aug 2024**
Mortgage Data Science Intern — *Troy, MI*

- Trained an XGBoost model in AWS SageMaker to predict loan prepayment risks, employing feature engineering on FICO scores, debt-to-income ratios, and regional economic indicators while addressing class imbalance with advanced sampling techniques.
- Conducted data cleaning, preprocessing, and exploratory data analysis (EDA) using **Python, NumPy, Pandas, Scikit-learn, and Seaborn** to improve data quality and uncover insightful patterns.
- Performed **SHAP** analysis and **LIME interpretation** on model predictions to identify key drivers—including interest rate differentials and borrower demographics—that informed targeted retention campaigns and reduced prepayment risks.
- Applied **Explainable AI (XAI)** techniques to validate model outcomes against legacy systems, generating transparent reports for stakeholders and ensuring alignment with regulatory requirements.
- Executed a comparative analysis between the Propensity Model and the Ice Model focusing on customer retention patterns, delivering actionable insights for strategic improvements.

- Developed interactive Power BI dashboards to visualize loan portfolio metrics such as lock volumes and error rates, empowering stakeholders with timely, data-driven insights.
- Collaborated with cross-functional teams to validate model outcomes and align retention strategies with overarching business objectives.

Cognizant **Dec 2020 - Jul 2023**
Data Analyst *India*

- Led the migration of **Teradata SQL workflows** to Google **Cloud Platform (GCP),** redesigning ETL pipelines to process millions of records of Kellogg's sales and inventory data.
- Optimized SQL queries for **BigQuery** and implemented parallel processing to reduce job runtime, enhancing **Apache Airflow** for workflow orchestration and Parquet for cost-effective storage.
- Reverse-engineered for complex **Teradata SQL** scripts and stored procedures to ensure accurate data lineage mapping during migration, resolving discrepancies in product inventory datasets and aligning transformed data with GRC Archer compliance frameworks.
- Developed **Python (Pandas, PySpark)** scripts to automate data validation, cleansing, and transformation across Teradata and GCP environments.
- Streamlined the conversion of legacy SQL logic into **Python-based ETL jobs**, ensuring seamless integration with GCP's serverless architecture.
- Built scalable SQL-driven pipelines in **GCP BigQuery** to ingest and transform unstructured data from **Kafka streams** and **Hadoop clusters**, enhancing data accessibility and processing efficiency
- Performed extensive Data Engineering using **Hadoop (MapReduce), Spark (PySpark), Kafka, and SQL-based ETL** processes for structured and unstructured data processing, improving data processing speed and reliability
- Crafted **ETL workflows** to successfully migrate and partition historical **Teradata** data into **GCP tables**, resulting in a significant reduction in query execution time through improved partitioning and clustering strategies.
- Collaborated with cross-functional teams to validate data integrity post-migration using **ANOVA and Chi-square tests**, ensuring data accuracy and reliability in the new system
- Automated legacy Teradata report generation by rewriting SQL-based business logic into **Python scripts**, integrating with **Power BI and Tableau** to deliver dynamic dashboards, which improved reporting efficiency and data visualization

Asriot Leo Tech. Pvt Ltd. **Jun 2020 - Jul 2020**
Research Intern *India*

- Contributed to a **CNN-based framework** for real-time COVID-19 detection, integrating thermal imaging data and symptom analysis to classify potential patients, with preprocessing pipelines built in OpenCV.
- Developed a novel deep learning framework for automated COVID-19 detection in images using convolutional neural networks (CNNs), addressing challenges in detecting among individuals.
- Preprocessed and augmented a dataset of 10,000+ images using **OpenCV and Python**, applying techniques like Gaussian filtering, histogram equalization, and synthetic data generation to enhance model robustness.
- Trained and fine-tuned models using **PyTorch and TensorFlow**, optimizing hyperparameters such as learning rate, batch size, and loss functions to achieve state-of-the-art performance.
- Validated model outputs against expert annotations, ensuring clinical relevance and improving diagnostic interpretability.
- Published findings as first author in ***Biomedical Signal Processing and Control (Elsevier)*** under the title **" *Framework for Real-Time Detection and Identification of possible patients of COVID-19 at public places*"**, detailing methodology, comparative analysis with existing models, and clinical implications.
- Conducted statistical analysis using **SciPy and Pandas** to pevaluate model performance metrics (sensitivity, specificity, AUC-ROC), demonstrating improvements over traditional computer vision approaches.

IIT Roorkee **Jun 2019 - Aug 2019**
Research Intern *India*

- Trained a multimodal summarization model using **PyTorch and TensorFlow** to generate human-like feedback from image-text pairs, aligning **ResNet-50** visual embeddings with **BERT-based** text representations.
- Preprocessed and augmented datasets of social media posts using Pandas and OpenCV, applying noise reduction and text normalization techniques to improve model input quality.
- Worked on the Summarization model, "Multimodal Affective Feedback", which generated feedback for input texts and the corresponding images, similar to how humans respond to multimodal data.
- To produce this type of feedback, a synthesis model has been trained on a large-scale dataset using ground-truth human comments and image-text input.
- Published research findings in IEEE on the application of deep learning for public health surveillance, detailing model architecture, training methodologies, and validation processes.
- Conducted exploratory data analysis on multimodal datasets, visualizing feature distributions and correlations using Matplotlib and Seaborn to guide model improvements.

EDUCATION

University of North Carolina at Charlotte **2024**
Master's, Data Science and Business Analytics

Graphic Era University, India **2021**
Bachelor's, Computer Science and Engineering